Vision- and Tension-Informed Inertial Identification and Formation Reconfiguration for Cooperative Aerial Transport Through Obstacles

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Abstract

Cooperative transport through spatial restrictions requires coordinated payload attitude, cable loading and aircraft formation. Visible shape determines occupied space, whereas hidden mass distribution determines the forces required to realize a configuration. This paper develops a vision- and tension-informed framework linking these quantities for rigid payloads and known obstacles. A markerless RGB-D scan establishes surface geometry and a common pose frame. Integral momentum observations and cable tensions identify mass, center of mass and inertia subject to a nonnegative mass-support model. Given smooth payload pose references, conic force optimization determines compliant support endpoints and selects feasible timing and lowering candidates under aperture, actuation and separation constraints. Six nominal closed-loop simulations yield mass errors below 0.02 percent and center-of-mass inertia errors of 5.46–7.87 percent. A five-arm ablation separates mass, center-of-mass and inertia information: for the four-aircraft rectangular task, peak attitude error decreases from 29.66 degrees with a mass-matched uniform model to 0.73 degrees with mass and center-of-mass identification and 0.71 degrees with full identification. Three- and four-aircraft teams transport rectangular and irregular payloads. An asymmetric forked payload completes a 28 m course with successive side-on, horizontal and opposite-side orientations, with peak position and attitude errors of 3.34 cm and 2.19 degrees. Transport-only prediction, sensor-seed repetitions, independent rigid-body replay and time-step refinement provide complementary numerical evaluations. The results connect sensed inertial moments to feasible support geometry during substantial payload reorientation.

Keywords: Cooperative aerial transport, Cable-suspended payload, Center of mass identification, Markerless perception, Payload reorientation, Formation reconfiguration

1. Introduction

Cable-suspended cooperative transport extends the manipulation capability of small unmanned aerial vehicles beyond the lifting capacity of one aircraft. Transport through infrastructure introduces an additional requirement: the suspended object and the complete supporting formation must change configuration to fit the available space. A lateral opening favors a reduced transverse span, whereas a low ceiling favors a reduced vertical span. An irregular load may also need to rotate substantially to orient a thin dimension toward an opening. Figure 1 presents the complete task: a scanned nonconvex payload and its supporting aircraft adopt different configurations while traversing a sequence of spatial restrictions. Here, payload reorientation means a change in its rigid-body attitude, whereas formation reconfiguration means a change in the relative aircraft endpoint positions.

Geometry alone cannot determine whether a desired configuration can be sustained. The load center of mass (CoM) generally differs from the centroid of its visible surface, especially when equipment or dense material is distributed unevenly inside the object. A misplaced CoM changes the moment produced by every cable and therefore changes the tension distribution required for a given attitude. Rotational inertia further affects the moments needed during attitude transitions. Because each cable force also determines an aircraft's direction and distance from its attachment, an inertial modeling error changes both tracking behavior and the formation's spatial envelope.

Several strands of research address these issues. Vision and inertial sensing have been used for cooperative cable transport [5], and nonlinear model predictive control coordinates transportation and load manipulation [6]. Arab et al. [8] study planning and distributed control for a nonuniform slung load. Load-distribution-based planning [9] connects the distribution of lifting effort to multilift trajectories. These studies establish that load distribution is a relevant planning variable rather than a consequence to be checked only after a centerline path has been selected.

Agile manipulation also requires the full supporting system to remain feasible. Sun et al. [1] demonstrate rapid cooperative manipulation using trajectory optimization and incremental nonlinear dynamic inversion. Their evaluations include substantial model mismatch and changing load properties. Thus,

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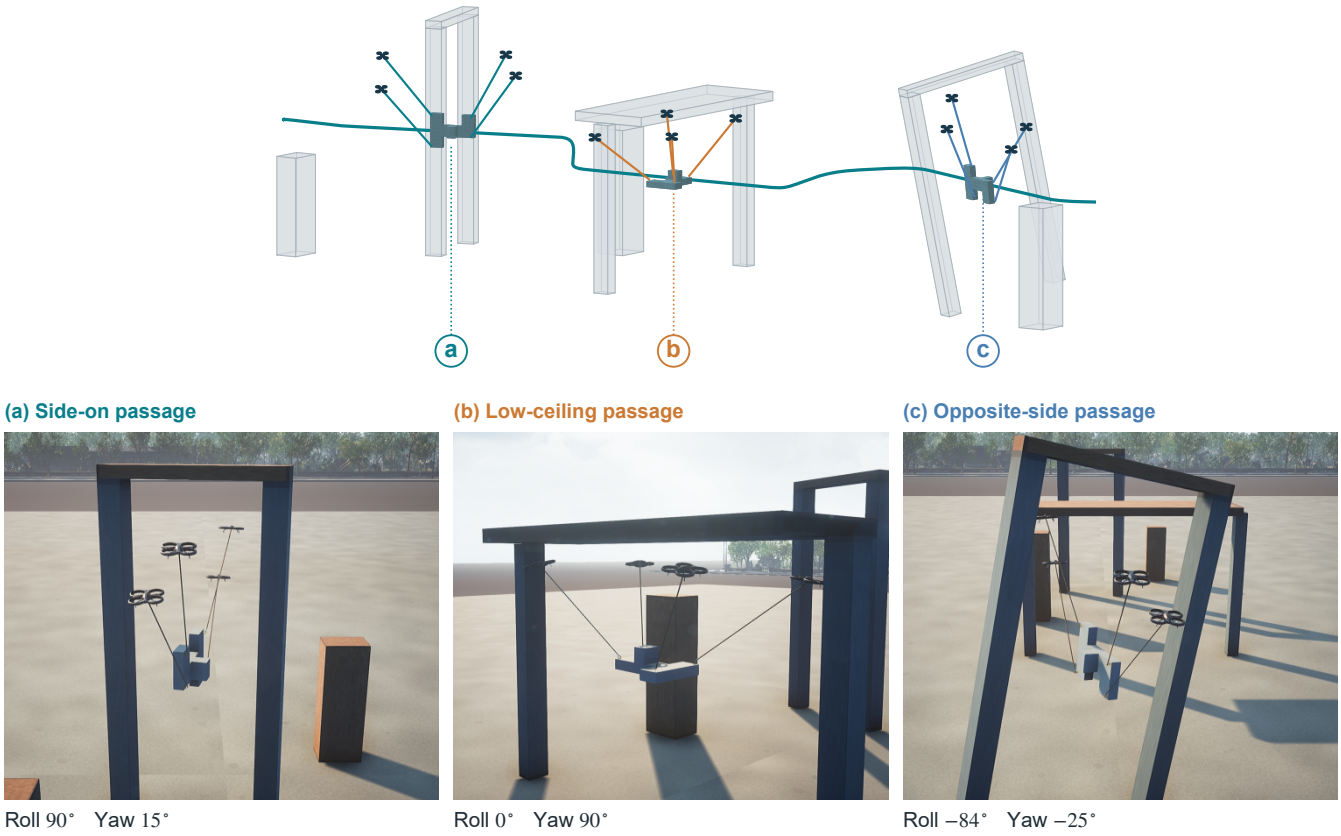


Figure 1: Cooperative transport through sequential spatial restrictions. The common-frame spatial view places three recorded payload and support configurations within the known obstacle course. Linked Unreal views show (a) side-on passage through the narrow opening, (b) a spread formation beneath the low ceiling, and (c) opposite-side passage through the tilted opening. Geometry and inertial observations supply the model used for force planning; the load attitude reference family is prescribed. The three configurations belong to one continuous closed-loop simulation.

agile payload rotation and mismatch-tolerant tracking are established capabilities. Control barrier functions have been used in cooperative manipulation trajectory planning [7], while PolyFly [14] uses polytopic descriptions for collision-free suspended transportation. These approaches motivate evaluating the geometry of aircraft and cables together with that of the payload. Learning-based direct motor control provides another route to cooperative transport [15]; the present study uses an explicit mechanical model so that estimated moments can be traced through force allocation to formation geometry.

Recent AST studies further clarify the planning context. Chai et al. [17] investigate time-optimal gap traversal for collaborative aerial transportation with experimental evaluation. Sun et al. [18] address local obstacle avoidance and trajectory tracking for multi-quadrotor transport. Chan et al. [19] formulate time-optimal planning of a cable-suspended payload quadcopter using spatial reformulation. The present objective is complementary: given an aperture-conditioned rigid-payload reference family, infer inertial moments from virtual geometry, tension and motion measurements, then propagate them through a compliant force-to-endpoint model. Timing is selected from a finite set; global time optimality and autonomous discovery of the attitude sequence are outside this formulation.

Inertial estimation supplies the complementary information. Petitti et al. [4] combine suspended-load inertial estimation with energy-efficient cooperative control. Physically consistent identification relates rigid-body inertial parameters to moments of a nonnegative mass distribution [3]. Heins and Schoellig [2] exploit uncertain inertial parameters and shape information in nonprehensile transportation. These results provide the basis for distinguishing external shape from internal mass properties. A surface scan supports spatial reasoning and constrains possible mass locations; force and motion observations are needed to identify the moments relevant to dynamics.

This paper addresses the connection between measured inertial information and feasible reconfiguration during obstacle passage. We consider a rigid object with calibrated cable attachment points, scanned before transport, and apertures whose geometry is known. The payload translation and attitude are parameterized by smooth, aperture-conditioned references. The planner determines the cable-force history and the resulting compliant endpoint trajectories, together with lowering and timing candidates. The aircraft then track these endpoints

through motor and rigid-body dynamics using delayed visual pose feedback. This formulation makes the influence of an identified CoM explicit: it alters the required wrench distribution and, through cable directions, the formation needed to realize the maneuver.

The contribution is an explicit connection from identified inertial moments to constrained support geometry, evaluated under a prescribed payload reference family. First, a sensing-to-configuration formulation combines markerless surface registration, synchronized tension observations and shape-supported integral inertial identification. It provides the mass moments used by the same force and geometry model during transportation. Second, a smooth conic force planner accounts for compliant cable endpoints, aperture halfspaces and aircraft actuation. For near-vertical load attitudes, sequential endpoint separation preserves distinct aircraft positions while the attachment geometry changes orientation. Third, controlled simulations examine three-aircraft and four-aircraft teams, compare identified and uniform-shape models, and demonstrate successive large-angle reorientations of a nonconvex payload through a multi-aperture course. Numerical refinement and an independent continuous command replay examine the underlying mechanical computation.

2. Transport model and problem formulation

2.1. Payload geometry and attachment coordinates

Consider $N \in \{3, 4\}$ quadrotors carrying a rigid payload through N individually attached cables. Let $p \in \mathbb{R}^3$ and $R \in \mathrm{SO}(3)$ denote the position and orientation of a reference frame fixed to the scanned object. This frame is determined geometrically and need not coincide with the CoM. Let b_i be the calibrated attachment coordinate in this frame, x_i the aircraft cable endpoint, and $\mathcal{S}$ the reconstructed payload surface. The world attachment is

$$a_i = p + Rb_i, \qquad d_i = x_i - a_i, \qquad q_i = \frac{d_i}{\|d_i\|}. \tag{1}$$

With tension $T_i \geq 0$, the cable force acting on the payload is $f_i = T_i q_i$. Noncollinear attachment geometry is required. Installed attachment coordinates are supplied by rigging calibration; they are not inferred from the object's geometric centroid.

The environment is represented by oriented obstacle boxes O_j. Collision evaluation considers the transformed surface $p + R\mathcal{S}$, each aircraft envelope and each complete cable segment $[a_i, x_i]$. The forked payload retains its open bays and protrusions in the collision model. A convex envelope is used separately for inertial support, as described in Section 3.

2.2. Rigid-body inertial parameters

Let m be the payload mass, c its body-frame CoM coordinate and J_O its inertia about the scan origin. The ten linear inertial parameters are

$$\theta = \begin{bmatrix} m & h^T & j^T \end{bmatrix}^T, \quad h = mc, \tag{2}$$

where j contains the six independent entries of the symmetric matrix J_O. The inertia about the CoM is

$$J_C = J_O - m(\,\|c\|^2 I - cc^T). \tag{3}$$

Let $v = \dot{p}$, ω and $\alpha = \dot{\omega}$ be world-frame linear velocity, angular velocity and angular acceleration, and let $g = [0, 0, -9.81]^T$ m/s^2. The cable wrench required about the scan origin satisfies

$$F = m(\ddot{p} - g) + ([\alpha]_\times + [\omega]_\times^2)Rh, \tag{4}$$

$$\tau_O = [Rh]_\times (\ddot{p} - g) + RJ_O R^T \alpha + \omega \times (RJ_O R^T \omega). \tag{5}$$

Consequently $w = [F^T, \tau_O^T]^T = Y_d(R, \omega, \ddot{p}, \alpha)\theta$ is linear in θ. For stacked cable forces $f = [f_1^T, \ldots, f_N^T]^T$, the same wrench is

$$G(R)f = w, \quad G(R) = \begin{bmatrix} I & \cdots & I \\ [Rb_1]_\times & \cdots & [Rb_N]_\times \end{bmatrix}. \tag{6}$$

The translation–rotation coupling in Eqs. (4)–(5) is retained during both identification and planning. In particular, a CoM offset produces a moment even during low-acceleration translation.

2.3. Compliant cables and aircraft dynamics

The cables are modeled as unilateral spring–damper elements with rest length ℓ_0, stiffness k_c and axial damping d_c. With $\delta_i = \|d_i\| - \ell_0$,

$$T_i = \begin{cases} \max(0, k_c\delta_i + d_c\dot{\delta}_i), & \delta_i > 0, \\ 0, & \delta_i \leq 0. \end{cases} \tag{7}$$

This law applies equal and opposite forces to the load and aircraft. The aircraft translational dynamics take the form

$$m_u\ddot{p}_i = u_i - f_i + m_u g - d_v\dot{p}_i, \tag{8}$$

where u_i is the rotor-generated thrust vector. Each aircraft has a free rotational state, and motor-site forces generate the corresponding body moments. A first-order motor response filters each thrust command before it acts on the body. The physical integration therefore includes cable stretch, aircraft attitude response and motor lag.

2.4. Planning objective

Let $p_r(t; \eta)$ and $R_r(t; \eta)$ define an aperture-conditioned reference family, where η selects the lowering and duration candidate. The problem is to find a smooth force sequence satisfying Eq. (6), cable loading limits and aperture conditions, and to convert that sequence into realizable aircraft endpoints. The complete endpoint and payload trajectories must satisfy the selected geometry and actuation checks. This finite candidate formulation specifies the motion family explicitly; obstacle geometry, attachment calibration and the acquired surface are inputs to the planner.

The coupling can be seen from the quasi-static endpoint map

$$x_i = a_i + \left(\ell_0 + \frac{\|f_i\|}{k_c}\right)\frac{f_i}{\|f_i\|}. \tag{9}$$

Changing $\hat{\theta}$ changes the wrench in Eq. (6), the admissible f_i, and the endpoint x_i. Thus internal mass information influences the spatial formation even for an unchanged external shape and payload reference.

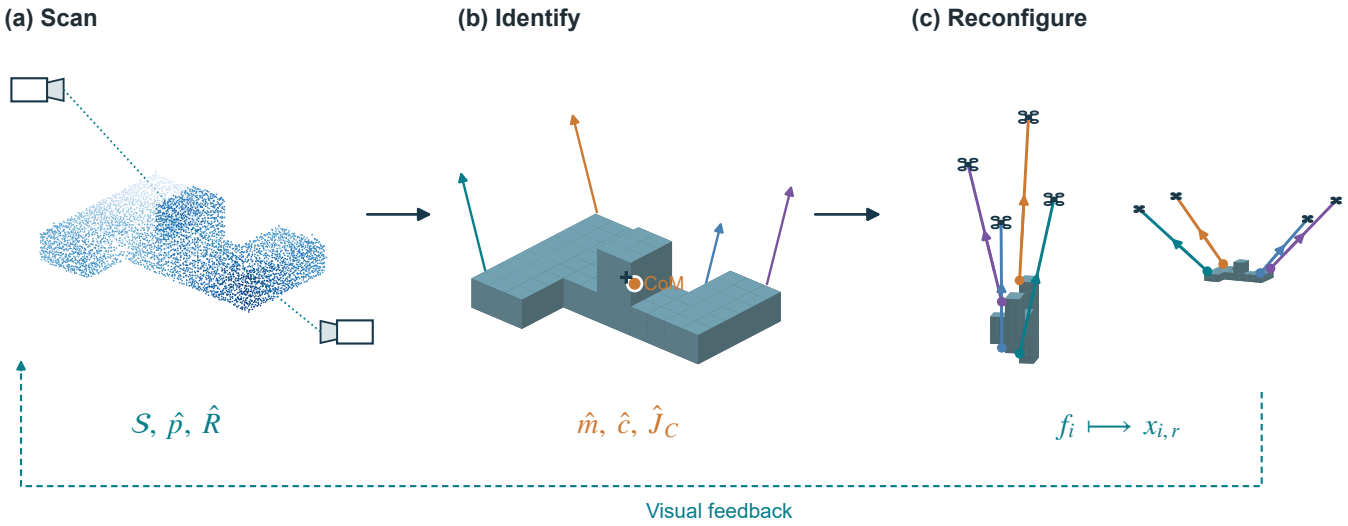


Figure 2: Mechanical schematic of the sensing-to-formation connection. (a) RGB-D observations provide the surface and registered pose. (b) Cable tensions and motion identify mass, CoM and inertia during preparation; the orange point marks the CoM and arrows indicate cable forces. (c) Constrained force allocation maps the identified moments to supporting endpoints during payload reorientation. Obstacle geometry, calibrated mounts and the smooth payload reference family are given inputs. The dashed path denotes transport pose feedback.

3. Vision and tension based inertial identification

3.1. Surface reconstruction and visual pose

The initial object scan is acquired in a known static installation fixture. Depth measurements are projected into calibrated coordinates and fused from twelve views. Voxel reduction produces a surface point set with up to 6000 points. The known fixture fixes the initial scan frame, including the orientation of a geometrically symmetric cuboid. Subsequent registration operates on the acquired surface rather than on a supplied ideal object mesh.

During transport, two virtual RGB-D views provide partial surface observations. A predicted search region follows the most recent registered pose, and geometric correspondence rejects points inconsistent with the reference surface. Point-cloud alignment follows the iterative registration principle [16]; temporal prediction initializes each local alignment. For the forked object, chromaticity statistics learned during the fixture scan filter background points while preserving thin prongs and cavity boundaries. The initial scan precedes rope installation, while transport images include the rigging. No fiducial marker or renderer object-ID mask is used.

The resulting visual observations are timestamped by exposure time. Tensions and aircraft endpoint telemetry are sampled at the same time. Their association is retained when image processing finishes later. The estimator therefore uses cable directions and load pose from a common physical instant rather than combining delayed geometry with current tensions. Nine consecutive observations are fitted by a local cubic polynomial, with derivatives assigned to the center observation. At the 10 Hz request rate this introduces an approximately 0.4 s fixed lag. Only observations already received enter an identification solve. This 0.4 s derivative lag applies to preparation-phase parameter estimation. During transport, the controller uses the most recent visual pose and a velocity-based prediction to the current control time; its reported sensor age is the exposure-to-control age, without the preparation derivative lag.

3.2. Integral momentum observations

An integral formulation avoids differentiating the visual trajectory twice. About a fixed world origin, payload linear and angular momenta are

$$P = mv + \omega \times Rh, \tag{10}$$

$$H = p \times P + (Rh) \times v + RJ_O R^T \omega. \tag{11}$$

Both are linear in θ, so define $[P^T, H^T]^T = Y_M \theta$. Gravity contributes the parameter-linear wrench

$$w_g = \begin{bmatrix} mg \\ (mp + Rh) \times g \end{bmatrix} = Y_g \theta. \tag{12}$$

The measured cable wrench about the same origin is

$$w_c = \begin{bmatrix} \sum_i T_i q_i \\ \sum_i a_i \times (T_i q_i) \end{bmatrix}. \tag{13}$$

For every complete interval $[t_a, t_b]$, integration gives

$$\left(Y_M(t_b) - Y_M(t_a) - \int_{t_a}^{t_b} Y_g \, dt \right) \theta = \int_{t_a}^{t_b} w_c \, dt. \tag{14}$$

The implementation uses trapezoidal integration over nonoverlapping 0.4 s intervals. Stacking the resulting equations produces $A\theta = b$. A translated fixed world origin near the first observation reduces unnecessary moment lever arms while preserving Eq. (14).

3.3. Shape-supported moment fitting

Let s_r be candidate mass-support locations inside the convex hull of the reconstructed surface. A nonnegative mass μ_r at each location contributes to mass, first moments and inertia. With a small isotropic cell-inertia term ϵI, the contributions are

$$m = \sum_r \mu_r, \quad h = \sum_r \mu_r s_r, \quad J_O = \sum_r \mu_r (\|s_r\|^2 I - s_r s_r^T + \epsilon I). \tag{15}$$

Write these relations compactly as $\theta = B\mu$. The estimate is obtained from

$$\begin{aligned} &\min_{\mu} \ \tfrac{1}{2}\|AB\mu - b\|^2 + \tfrac{\lambda}{2}\|\mu\|^2 \\ &\text{s.t. } \mu \geq 0, \qquad m_{\min} \leq \mathbf{1}^T\mu \leq m_{\max}. \end{aligned} \tag{16}$$

This is a convex quadratic conic program [13]. The nominal fit minimizes the unwhitened residual in SI units. Linear- and angular-impulse residuals are reported separately. To assess the relative scaling of their different units, the supplementary analysis rescales moment rows by $1/L$ for $L \in \{0.25, 0.5, 1, 2\}$ m and refits the same observations; this is a unit-weighting sensitivity test, not covariance whitening. The support spacing is 0.08 m, $\epsilon = 5 \times 10^{-5}$ m^2, $\lambda = 10^{-8}$, and the prescribed mass interval is 0.7–1.5 kg. A limited set of hull vertices supplements the interior grid. Nonnegative support guarantees moments realizable by mass within the scanned convex envelope, following the mass-distribution interpretation of inertial parameters [3].

For a nonconvex object, this hull is a conservative envelope of possible mass locations. It serves the moment estimation problem, whereas the reconstructed nonconvex surface serves collision evaluation. The envelope therefore imposes moment consistency and a spatial outer bound, rather than identifying a material-only density. The weights in Eq. (16) are one representation of the fitted moments: many different internal distributions can generate the same ten parameters. The estimated quantities passed to the planner are $\hat{m}$, $\hat{c}$ and $\hat{J}_O$.

An activation test checks the measurement geometry before using these moments. The normalized observation matrix must have rank ten, condition number at most 1000 and at least eight complete integral windows. The minimum eigenvalue of the mean $\omega\omega^T$ matrix must exceed 10^{-4} s^{-2}. Measured tension must exceed its 0.01 N error bound, and the central second-moment matrix $\frac{1}{2}\,\mathrm{tr}(J_C)I - J_C$ must be positive definite. These conditions connect activation to angular excitation, valid tension measurements and physical moment consistency.

4. Coupled force and formation planning

4.1. Aperture-conditioned motion references

The inputs are the reconstructed surface, calibrated attachments, known obstacle boxes, fitted inertial parameters and a prescribed smooth payload reference family. The continuous optimization variables are the three components of every cable force at each time node. Duration and lowering are selected by ordered finite candidate search. Payload position and attitude follow the selected reference; aircraft endpoints are derived from optimized forces and cable stretch. Thus force allocation and support formation adapt to the measured load, while the obstacle-crossing attitude sequence is supplied by the reference family.

The load reference is represented by smooth translation and attitude functions over a selected duration D. The basic experiment translates the payload by 8 m while moving toward a terminal roll, pitch and yaw of 25°, 10° and 30°. The reference includes a smooth lowering window under the beam. Candidate durations are 20, 22, 25 and 30 s; candidate lowering values are evaluated in a fixed order.

For sequential apertures, a piecewise quintic centerline connects their approach and exit neighborhoods. Smooth activation functions blend the associated attitude references. In the large-angle course the sequence is horizontal, 90° roll at the narrow aperture, horizontal with 90° yaw under the ceiling, −85° roll at the tilted aperture, and return to the nominal attitude. The references vary during forward motion. Their activation regions extend across the occupied aperture neighborhood, so a load transition is not released while supporting aircraft still occupy the frame. Duration and lowering candidates are evaluated using the same force and endpoint checks.

4.2. Smooth conic cable-force optimization

At time nodes t_j, let $f_j \in \mathbb{R}^{3N}$ contain the load cable forces, and let f_j^0 be a preferred outward preload. For a chosen motion candidate, the force planner minimizes

$$\begin{aligned} \mathcal{J}(f) = &\frac{0.02}{2}\left\|f - f^0\right\|^2 + \frac{0.04}{2}\|D_1 f\|^2 \\ &+ \frac{0.20}{2}\|D_2 f\|^2 + \frac{0.04}{2}\|D_3 f\|^2, \end{aligned} \tag{17}$$

where D_1, D_2, D_3 are time-scaled first, second and third finite-difference operators. Force equilibrium is imposed at every node:

$$G(R_r(t_j))f_j = Y_d(t_j)\hat{\theta}. \tag{18}$$

The formulation includes upward force, tension and nominal thrust cones,

$$f_{ij,z} \geq 0.55, \quad \left\|f_{ij}\right\| \leq 8.2, \quad \left\|f_{ij} + m_u g_0 e_z\right\| \leq 29.5, \tag{19}$$

with forces in newtons, $e_z = [0, 0, 1]^T$, and $g_0 = 9.81$ m/s^2 denoting the positive gravity magnitude. In the thrust cone, $m_u g_0 e_z = -m_u g$ with the vector convention of Section 2. The 29.5 N cone limits nominal total thrust; four 7.5 N motor limits permit 30 N total before moment allocation. Dynamic endpoint verification applies a tighter 28 N nominal limit. Componentwise limits of 8 N/s and 25 N/s^2 are placed on the first and second differences. Near-zero first differences at the endpoints reduce force transients when joining the preparation and transport phases. The initial force is projected onto the first wrench equality to maintain continuity.

An outward preload prevents the formation from collapsing above the object. For large roll angles, horizontal aircraft sectors are defined using a horizontal projection of the payload longitudinal axis and its orthogonal horizontal direction. The

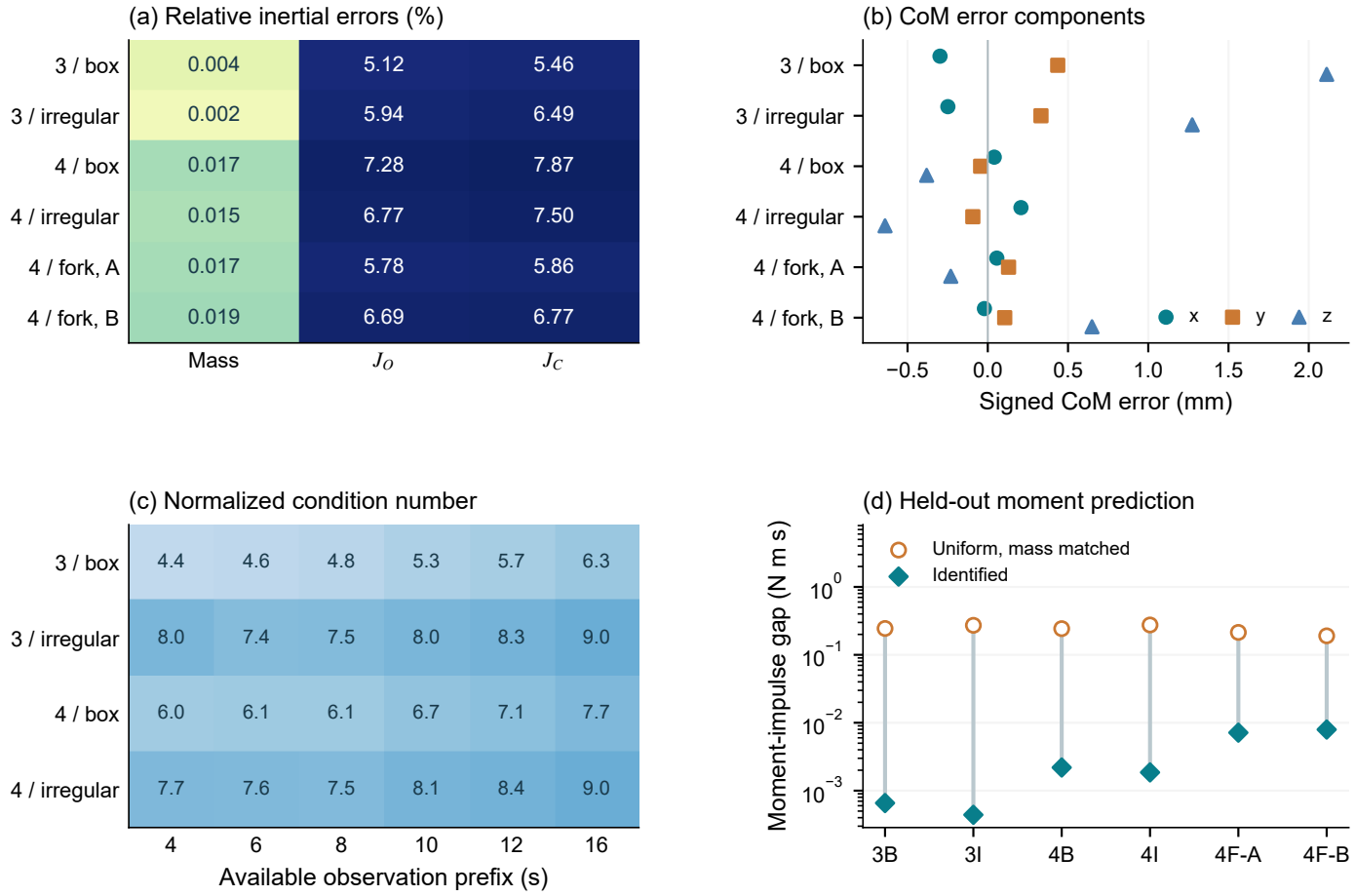


Figure 3: Inertial accuracy and predictive evaluation. (a) Relative mass and inertia errors in six recorded executions; inertia errors use the Frobenius norm. (b) Signed CoM errors in the scan frame; circles, squares and triangles denote the x, y and z components. (c) Condition number of the column-normalized integral observation matrix using only data available by each prefix; all displayed systems have rank ten. (d) RMS difference from the true-inertia moment-impulse prediction on transport observations excluded from fitting, comparing identified and mass-matched uniform models. Fork A and B denote the moderate- and large-rotation courses. Heatmap colors use logarithmic scales; cell annotations give the numerical values.

sectors retain the original attachment ordering when the attachment plane becomes nearly vertical. Side-force conditions are relaxed with increasing roll to allow the force couples required by the upper and lower attachments. The subsequent endpoint and cable checks determine acceptance.

4.3. Aperture constraints and compliant endpoints

For an aperture halfspace $n^T x_i \le \beta$ and known attachment a_i, define the endpoint allowance $c_i = \beta - n^T a_i > 0$. A conservative linear force condition used in the conic problem is

$$\left(n - \frac{c_i}{\ell_{\max}} v_i\right)^T f_i \le 0, \tag{20}$$

where v_i is a prescribed unit direction and $\ell_{\max} = 2.05$ m bounds the planned cable length. Since $v_i^T f_i \le \|f_i\|$, this condition implies $n^T f_i \le (c_i/\ell_{\max})\|f_i\|$. Together with $\ell_i \le \ell_{\max}$, it bounds the nonlinear endpoint displacement toward the halfspace. Height limits are expressed by related outward/upward force inequalities. Full nonlinear geometry is checked after solving.

For a smooth force magnitude $T_i(t) = \|f_i(t)\|$, stretch is computed from

$$d_c\dot{\delta}_i + k_c\delta_i = T_i(t), \qquad x_{i,r} = p_r + R_r b_i + (\ell_0 + \delta_i)\frac{f_i}{T_i}. \tag{21}$$

The stretch dynamics are propagated at 5 ms resolution, and clamped cubic splines supply the endpoint position, velocity and acceleration. This step retains axial damping instead of instantaneously changing cable length whenever tension changes. Aircraft acceleration, dynamic thrust and tilt are evaluated on the resulting endpoint curves, supplementing the nominal cones in Eq. (19).

4.4. Separation during large-angle reorientation

Near-vertical payload orientations can bring different endpoint directions close together even when the attachments themselves are well separated. We therefore add sequential linearized endpoint separation. At a preceding force solution $\bar{f}_i$, the static elastic map in Eq. (9) has Jacobian

$$J_i = \frac{\ell_0}{\|\bar{f}_i\|}\left(I - \bar{q}_i\bar{q}_i^T\right) + \frac{1}{k_c}I. \tag{22}$$

For the previous endpoint difference direction $n_{i\ell}$, impose

$$n_{i\ell}^T \left[\bar{x}_i + J_i(f_i - \bar{f}_i) - \bar{x}_\ell - J_\ell(f_\ell - \bar{f}_\ell) \right] \geq d_{\text{lin}}. \tag{23}$$

Three successive conic updates use $d_{\text{lin}} = 0.72$ m. The final damped endpoint trajectory is then evaluated nonlinearly. Its acceptance distance is 0.52 m; the linearization distance is an optimization reference rather than a guaranteed executed separation. This distinction matters because damping, spline interpolation and tracking alter the static linearization geometry.

Feasibility is evaluated at three distinct levels. The conic solve enforces nodal wrench equalities and force cones. Candidate verification evaluates damped, interpolated endpoints and complete geometric occupancy. Closed-loop execution is assessed using the measured tracking, loading and actuation limits in Section 5. The 0.72 m separation linearization reference encourages spacing during optimization; the nonlinear candidate gate is 0.52 m. Execution reports aircraft center distance separately from clearance after subtracting the collision envelope. Preparation transients are also included in execution tension extrema; they are not subject to the transport-only force reference bound.

The course uses 121 force nodes. Final evaluation samples the full acquired surface, aircraft envelopes and complete cable segments at 20 ms intervals, with actuation evaluation on the finer endpoint curve. A candidate is accepted only after its nonlinear force, geometry, speed and acceleration checks pass. Accepted course references satisfy aircraft speed below 2.5 m/s, reference acceleration below 2.75 m/s^2, tension between 1 and 8.25 N, dynamic thrust below 28 N and tilt below 37°. The single-beam reference acceleration budget is 2.25 m/s^2. These budgets leave room for tracking corrections in the execution loop.

4.5. Closed-loop execution

At each control instant, a delayed registered load pose is propagated using its available velocity estimate and combined with current aircraft telemetry. A low-pass payload pose correction shifts the endpoint references. Wider corner mounts use a lower visual pose gain to keep delayed rotational corrections below the coupled load-swing response. The aircraft command is

$$u_i = m_u \left(\ddot{x}_{i,r} + K_p(x_{i,r} + \Delta_i - x_i) + K_v(\dot{x}_{i,r} - \dot{x}_i) - g \right) + f_i + d_v \dot{x}_i, \tag{24}$$

where $K_p = 18I$ s^{-2} and $K_v = 7I$ s^{-1}, Δ_i is the filtered pose correction, and requested acceleration is capped at 4.5 m/s^2. A geometric attitude controller [10] aligns each aircraft's thrust direction with this command, followed by allocation to four filtered motor inputs.

Image registration runs in its own worker; the revised implementation isolates that worker in a process to separate rendering/registration from control scheduling. At most one image request is outstanding, and the maximum permitted age of the latest valid exposure is 0.35 s. Identification and whole-segment planning run in a separate process; the aircraft remain under preparation/hover control while these computations finish. After the candidate is ready, the transport clock starts and the 50 Hz feedback loop follows its reference. This arrangement assigns explicit computation and observation times to the complete sensing–planning–execution chain.

Table 1: Dynamic and sensing parameters. Noise bounds apply to the corresponding virtual measurements.

Quantity	Value
Aircraft mass	1.0 kg
Aircraft inertia diagonal	$[.009, .009, .016]$ kg m^2
Cable rest length	2.0 m
Cable stiffness / damping	180 N/m / 5 N s/m
Maximum thrust per motor	7.5 N
Motor response time constant	25 ms
Aircraft linear drag coefficient	0.2 N s/m
Local ICP RMSE / inlier gate	18 mm / 0.65
Correspondence gate	80 mm
Control / RGB-D request rate	50 Hz / 10 Hz
RGB-D image resolution	256×192 pixels per view
Depth noise standard deviation	1 mm
Tension noise	Uniform ± 0.01 N
Aircraft position noise	Uniform ± 0.5 mm
Preparation motion	16 s
Physical integration step	0.5 ms

5. Simulation experiments

5.1. Experimental setup and evaluation criteria

The dynamic simulator integrates the payload and every aircraft as free rigid bodies in MuJoCo [11]. Individual rotor-site forces generate aircraft translation and rotation, and the cable law in Eq. (7) generates the load reactions. Table 1 summarizes the model and sensing settings. The default physical integration step is 0.5 ms, and motor commands are updated at 50 Hz. Computation is executed on a workstation with an AMD Ryzen 9 8945HX processor and an NVIDIA GeForce RTX 5070 Ti Laptop GPU.

The nominal task group N comprises rectangular and nonconvex stepped payloads, each carried by three and four aircraft, as illustrated in Fig. 4. Each configuration is paired with a nominal uniform-shape control model, giving eight attempts (N3B, N3I, N4B and N4I denote the identified-model three/four-aircraft box/irregular executions). A separate group A contains the twenty initial five-arm ablations and the original twenty sensor-seed attempts; group B contains the process-isolated sensor experiments introduced below. Every record has an experiment path, sensor seed, fixed ballast seed, configuration and source-hash manifest in the public experiment identity index. Within each nominal pair, the object, hidden ballast, sensor seed, preparation procedure and planning rules are held fixed. The uniform model distributes a nominal 1.05 kg over interior scan-support locations, while the identification model uses Eq. (16). The comparison therefore evaluates the inertial information supplied to planning and feedback under the same acquired external geometry.

Two further nominal courses carry a forked payload through sequential restrictions. The object measures $1.21 \times 0.90 \times 0.40$

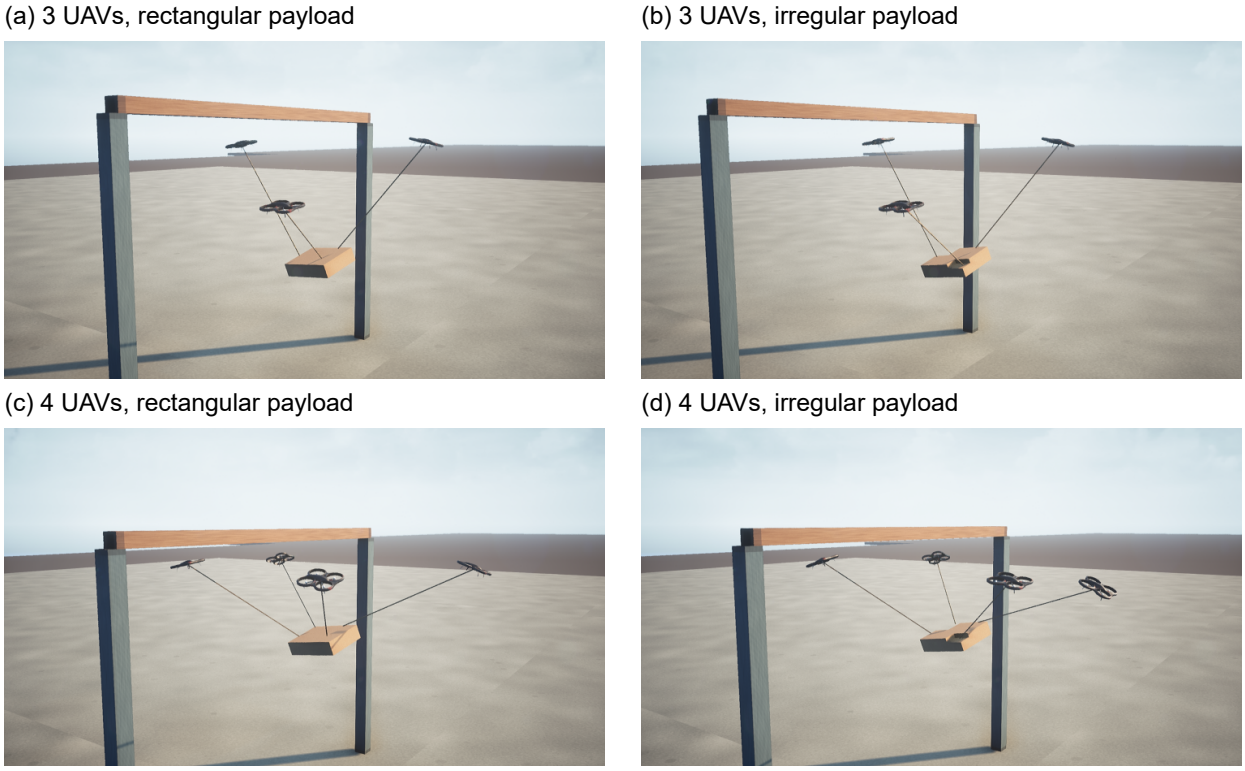


Figure 4: Team-size and shape configurations in the base experiments. (a),(b) Three UAVs carrying rectangular and irregular payloads. (c),(d) The corresponding four-UAV configurations. All panels use original Unreal images of the recorded dynamic executions.

m and has unequal side arms, open bays and a raised boss. Four mounts are installed at material-supported outer corners. Fixed padeyes, rotating shackles and rope terminations share the same attachment geometry as the force model. Mount mass and inertia are included in the load properties. The effective 2 m cable includes the 60 mm rigid termination, with the force application point at its pin center. These details permit large rotations while maintaining consistent visual and mechanical attachments.

Execution acceptance requires completed transport, peak position error at most 0.10 m, peak rotation error at most 5°, and tensions strictly between 0.5 and 9 N. Additional limits are 40° aircraft tilt, 7.5 N per motor, 2.6 m/s aircraft speed and 6 m/s^2 aircraft acceleration. Rotation error is the magnitude of the logarithm of RR_r^T. Rigid-body contacts are monitored at physical substeps. Complete cable-segment checks use the 50 Hz records; the course clearance analysis inflates segment radius by the termination width and half a sampling interval of endpoint motion. Corresponding obstacle and load-body checks use the complete saved segment, rather than only cable endpoints.

The RGB-D inputs used for identification and feedback are rendered from the dynamic state and returned through the image-processing interface. Separately, Unreal Engine 5.2 and the CityPark Showcase environment produce the presentation images and videos. The distinction follows the useful separation between visual and physical simulation [12]: in this study numerical performance is measured from the MuJoCo closed loop, and Unreal displays those recorded states. The video output is 1920 × 1080 at 25 frames/s with one-to-one physical time. No display interpolation changes the recorded motion.

5.2. Complete inertial identification and held-out prediction

Across the four base configurations and two forked-payload executions, mass relative error is below 0.02%, and the CoM-inertia Frobenius relative error is 5.46–7.87%. True and estimated mass, three-axis CoM, all six origin-inertia entries and principal moments are provided in the public numerical records. Figure 3(a),(b) therefore evaluates the complete ten-parameter model rather than using CoM accuracy as a proxy for inertia accuracy. Figure 3 compares the models used for transportation. The identified CoM errors range from 0.39 to 2.18 mm across the four configurations. In contrast, the uniform-shape models differ from the actual CoM by approximately 103–117 mm. The external shape and installation coordinates are available to both models, so this difference concerns internal mass moments rather than surface acquisition.

Figure 3(c) evaluates observation geometry at causal prefixes from 4 to 16 s. The normalized systems have rank ten, with condition numbers of approximately 4.4–9.0. At a 6 s prefix, the original activation criteria are satisfied in all four base cases. The prefix calculations share the original observation streams and quantify information availability during preparation.

For predictive evaluation, complete 0.4 s integral windows are formed from transport-only observations, excluding the first and last second of transport. These samples do not enter the preparation fit. Figure 3(d) compares each model's predicted

moment impulse against the true-inertia prediction on the same held-out regressor. Matching the uniform model's total mass isolates its remaining first- and second-moment errors. Table 3 additionally evaluates $A\theta - b$ against the measured cable impulse for oracle, identified and mass-matched uniform models. The oracle residual includes the image-motion, tension and quadrature error of this same simulated observation chain; it supplies a reference residual floor rather than independent physical validation.

The completed four-aircraft rectangular pair provides a direct tracking comparison. Its uniform model has 102.75 mm CoM error and 22.05° peak attitude error. With the identified model, these values are 0.39 mm and 0.66°. Peak translation error decreases from 8.19 to 2.76 cm. Thus the attitude error is reduced by approximately 97% for this pair. The change has a mechanical explanation: a misplaced CoM creates an incorrect gravity moment in Eq. (5), which produces inappropriate cable-force distribution during rotation and load lowering.

The identified and oracle measured residuals are close, whereas the uniform-moment discrepancy is particularly pronounced in four-aircraft transportation. This comparison includes the observation-chain residual floor. The small nominal mass errors arise under bounded virtual measurement noise and the shared rigid-body model; they do not specify the accuracy of a physical RGB-D/tension installation.

5.3. Team size and payload geometry

All four identified-model configurations satisfy the execution criteria. Table 2 reports their individual results rather than treating them as repeated samples of one configuration. Peak position errors are 2.62–4.10 cm, and peak attitude errors are 0.66–2.64°. The three-aircraft irregular case selects 22 s from the candidate duration set; the other three use 20 s. Their peak payload speeds are 0.90–0.99 m/s, and average speeds within the central passage region are 0.84–0.93 m/s.

Figure 6 shows cable loading during transportation. Across preparation and transport, tensions remain between 1.03 and 8.69 N. Aircraft, payload and cable-obstacle contacts are absent, and none of the identified-model runs saturates its motor commands. The four-aircraft rectangular formation reduces the vertical endpoint-to-payload envelope from a maximum of 1.94 m to a minimum of 1.09 m during the maneuver, while the endpoint lateral span varies between 1.86 and 3.38 m. The formation therefore changes the occupied vertical space by spreading the support directions as the payload passes beneath the beam.

The uniform-model outcomes in Table 4 complement the tracking example. Three configurations do not obtain an accepted candidate. The completed four-aircraft rectangular execution exceeds the attitude and aircraft acceleration limits. This comparison shows that the scanned outline alone does not provide the same task model as the outline augmented by measured inertial moments. Candidate rejection here is a result of the specified finite search, not a claim of global infeasibility of the physical task.

5.4. Controlled ablations and sensor-seed repetitions

Groups N and A are separate executions, not repeated labels for the same trajectories. Although both use sensor and ballast seed 2501, asynchronous image delivery and preparation/planning waits change the exposure times and transport initial states. This explains the N/A attitude peaks of 0.66/0.71° for the four-aircraft box and 1.29/1.06° for the four-aircraft irregular load. The uniform results also differ because group N assumes 1.05 kg while group A matches the true total mass. The public identity index maps every table entry to its source record.

The controlled comparison uses five inertial-information levels on each of the four base configurations: a uniform-shape model with true total mass; identified mass with uniform-shape CoM and central inertia; identified mass and CoM with mass-scaled uniform-shape central inertia; full identification; and true inertial parameters. All arms use the same acquired-geometry procedure, fixed ballast, sensor seed, preparation, reference family, candidate budget and feedback gains. The true-mass arm isolates errors in the first and second moments without a mass mismatch. The oracle is an information reference, rather than a competing planner.

Table 5 distinguishes candidate availability from completed-transport accuracy. Both uniform-moment arms produce a candidate only for the four-aircraft rectangular configuration; its peak attitude errors are 29.66° and 29.44°, respectively. All four full-identification and four true-inertia attempts satisfy the execution criteria. Three mass–CoM attempts complete and satisfy the criteria; the remaining acquisition abort is reported separately. In the four-aircraft rectangular case, their attitude peaks are 0.731°, 0.712° and 0.678°. The dominant tracking gain in this task therefore follows CoM correction. The incremental effect of identifying central inertia is smaller and depends on the configuration; the three-aircraft rectangular mass–CoM arm, for example, has a slightly smaller attitude peak than its full-identification counterpart. Complete-inertia accuracy is assessed separately by Fig. 3 and transport-only prediction.

The three-aircraft irregular mass–CoM initial attempt stops on the image-age gate before transport and is retained as A, rather than classified as planning rejection. A separately labeled acquisition retry completes the task with a 2.54° attitude peak. It does not replace the original table entry.

Twenty additional initial attempts use independent virtual sensor seeds for the fixed four-aircraft rectangular task. All 20 obtain an initial plan; 16 complete and satisfy the execution criteria, while 4 stop on the image-age gate. The all-attempt completion-and-acceptance rate is 16/20 (80%; Wilson 95% interval 58.4–91.9%). Among the 16 complete transports, peak attitude error is 0.606–0.851°, and its median is 0.725°. Transport-only position RMS has median 1.96 cm and p95 1.97 cm; attitude RMS has median 0.427° and p95 0.488°. Initial planning time among complete transports has median 1.84 s and p95 1.89 s. These distributions are conditional on completed transport, and the initial attempts remain the statistical cohort even when a separately labeled diagnostic retry is available. The repeated task fixes shape and ballast;

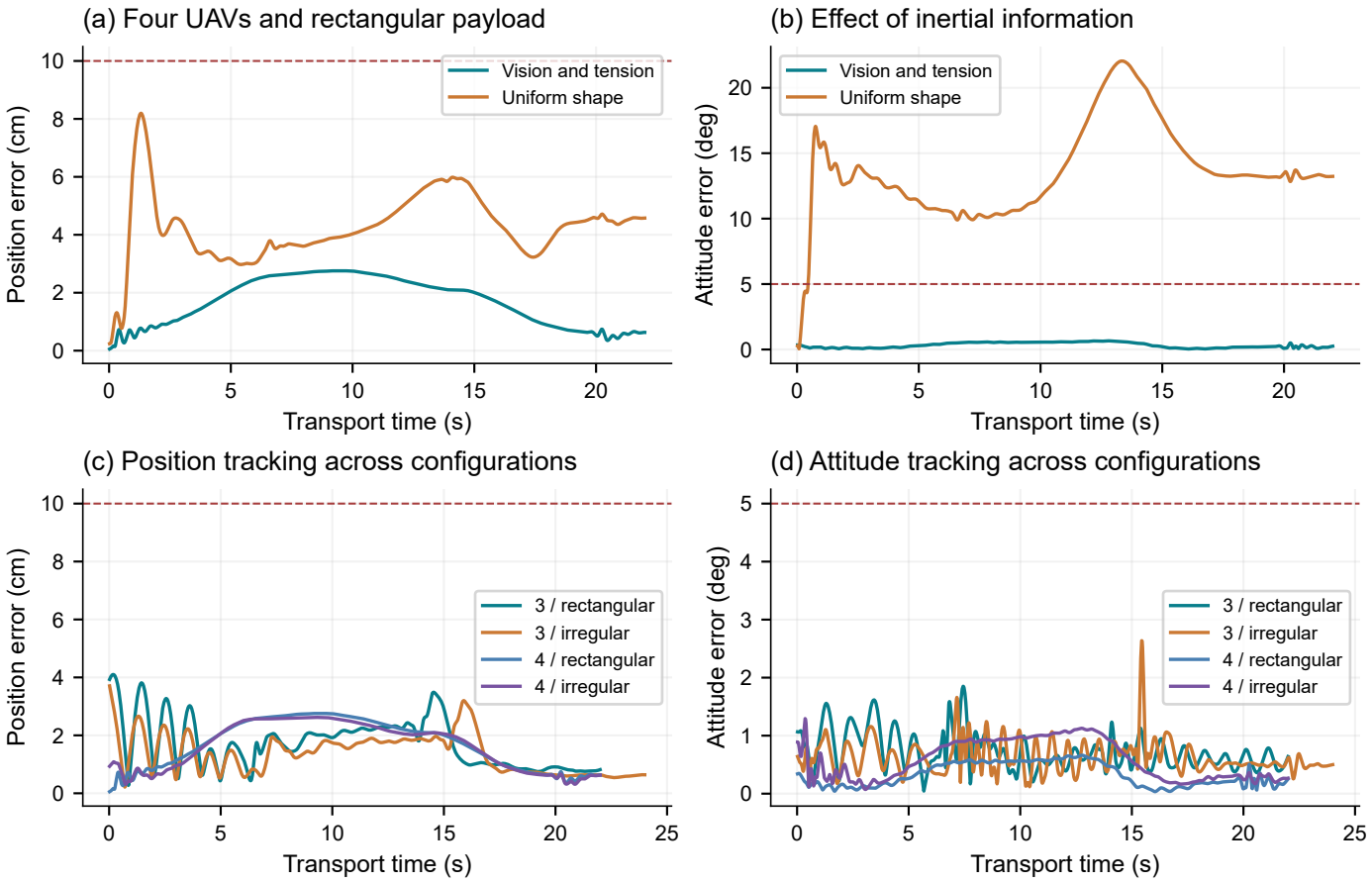


Figure 5: Closed-loop tracking in the base experiments. (a),(b) Paired four-aircraft rectangular-payload executions with uniform-shape and identified inertial models. (c),(d) Identified-model position and rotation errors for all four team/shape configurations. Dashed lines mark the execution thresholds. Time is relative to each transport start.

Table 2: Group N nominal transport results with identified moments (rows N3B, N3I, N4B, N4I). Each row is one executed team/shape configuration. Tension extrema include the complete preparation and transport at physical substeps.

UAVs	Payload	Transport (s)	CoM error (mm)	Peak position (cm)	Peak attitude (°)	Tension (N)
3	Rectangular	20	2.18	4.10	1.85	1.08–8.38
3	Irregular	22	1.34	3.71	2.64	1.20–7.55
4	Rectangular	20	0.39	2.76	0.66	1.63–8.69
4	Irregular	20	0.68	2.62	1.29	1.03–8.61

its seed distribution concerns sensing and acquisition timing under that configuration.

The process-isolated group B repeats exactly seeds 27000–27019 once each, retaining group A unchanged. All 20 new initial attempts complete and satisfy the common execution gates (20/20; Wilson 95% interval 83.9–100%). Their transport-only position RMS median/p95 is 1.96/1.97 cm, and attitude RMS is 0.397/0.431°; peak attitude error is 0.587–0.759°. Relative mass error has median 0.0074% and p95 0.0165% under the declared virtual sensor model. Across these executions, the minimum aircraft-envelope obstacle clearance is 0.0408–0.0413 m. These results describe a fixed task with new sensor executions, rather than a cross-object reliability guarantee.

The revised image pipeline runs rendering and registration in an isolated process, with one outstanding frame and the unchanged 0.35 s maximum age of the latest valid exposure. Per-frame wall-clock stamps give p95 queue/dispatch, image-processing and return/control-read times of 5.33, 69.97 and 26.91 ms; exposure-to-read p95/max is 85.49/111.41 ms. Their separate quantiles are not additive. The four original aborts occurred at physical times 19.42, 34.06, 36.66 and 39.58 s. Original logs retain processing and exposure/receipt times, but do not support retrospective queue/IPC decomposition. The supplementary timing figure presents a failure timeline and both complete cohorts. If the age gate expires, the simulated execution is terminated and counted as incomplete; no blind continuation or automatic retry is included.

Table 3: Transport-only measured cable-impulse residuals $A\theta - b$ on the same held-out windows. U denotes the true-mass uniform model; ID denotes full identification. RMS is taken over all windows and three components, separately for force and moment. The oracle uses the simulator's true moments.

	Force impulse RMS (N s)			Moment impulse RMS (N m s)		
Case	Oracle	ID	U	Oracle	ID	U
N3B	0.0487	0.0487	0.0493	0.2269	0.2269	0.3337
N3I	0.0431	0.0431	0.0428	0.2235	0.2235	0.3506
N4B	0.0057	0.0057	0.0058	0.0194	0.0192	0.2411
N4I	0.0047	0.0047	0.0048	0.0135	0.0133	0.2732
Fork A	0.0278	0.0278	0.0283	0.3917	0.3918	0.4476
Fork B	0.0224	0.0223	0.0214	0.2054	0.2051	0.2646

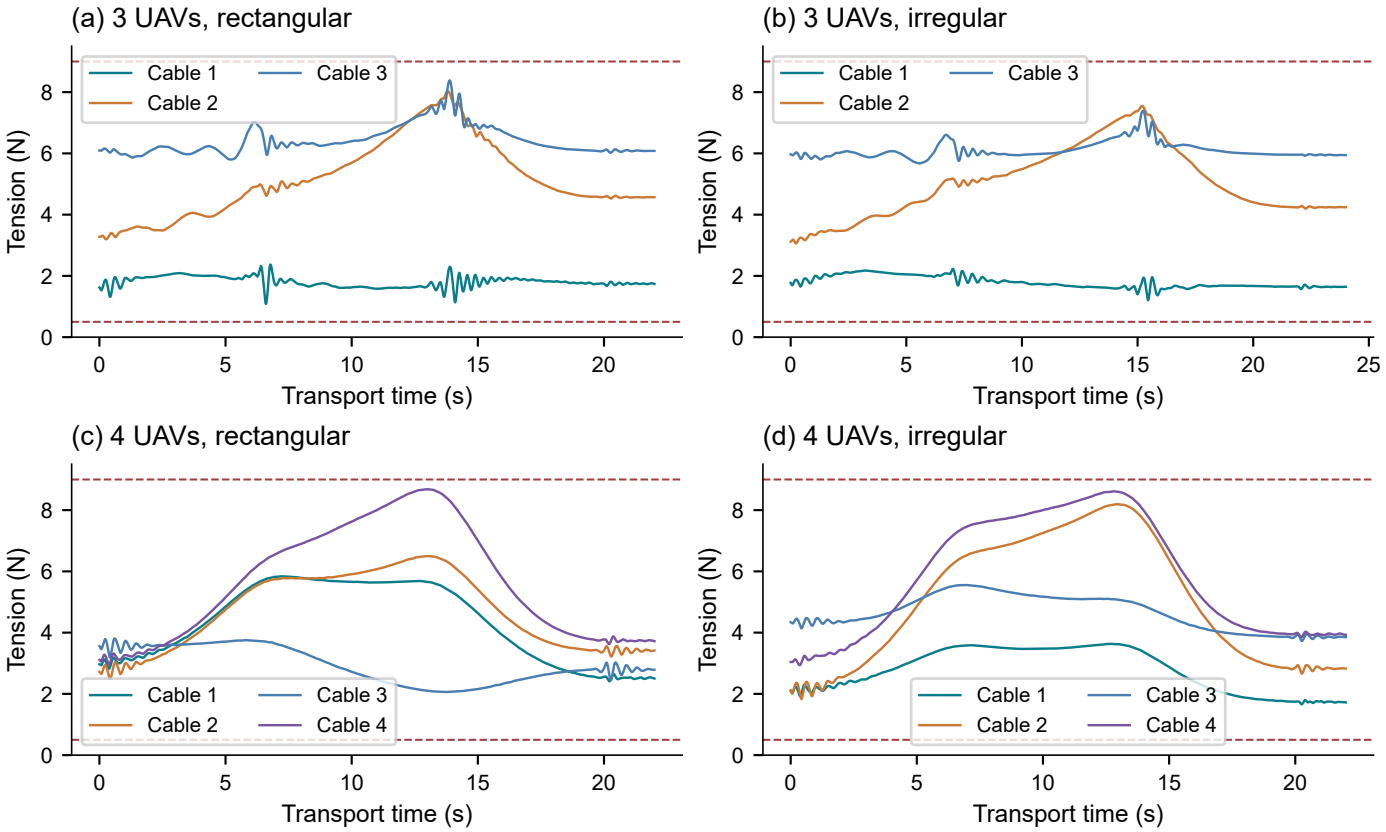


Figure 6: Cable tensions during transport for the four identified-model configurations. Curves show 50 Hz saved states; dashed bounds are the 0.5–9 N execution limits. The extrema in Table 2 additionally cover every physical substep and the preparation phase.

Table 4: Group N nominal uniform (1.05 kg) and identified-model outcomes under the same task rules. "Planning rejection" denotes that the candidate set did not produce an accepted force/geometry/actuation trajectory for that model.

UAVs	Payload	Uniform CoM error (mm)	Uniform-model outcome	Identified outcome
3	Rectangular	103.35	Planning rejection	Accepted
3	Irregular	115.31	Planning rejection	Accepted
4	Rectangular	102.75	Attitude limit exceeded	Accepted
4	Irregular	116.74	Planning rejection	Accepted

5.5. From inertial parameters to support geometry

To isolate the mechanical link, each group N reference is frozen at its recorded duration, lowering and initial payload pose. All allocations use that same reference and acquired scan, rather than selecting different motions for different models. A controlled 10 mm shift of the identified CoM along the scan x axis retains its mass and central inertia. Table 6 reports the change in optimized cable forces and compliant endpoints. This is a planning sensitivity comparison, not another set of executed flights.

The 10 mm parameter change produces 0.128–0.230 N RMS force changes and 3.86–13.29 cm RMS endpoint

Table 5: Group A five-arm controlled ablation (rows A3B, A3I, A4B, A4I). The uniform arm matches the true mass, unlike the 1.05 kg nominal uniform in Table 4. Entries give peak attitude error (°) for completed executions. R denotes candidate planning rejection; A denotes an image-age abort before transport. Asterisks mark completed executions outside the common acceptance criteria. Each cell is one initial attempt, with all attempt records retained.

UAVs	Payload	True mass, uniform	Mass only	Mass + CoM	Full identification	True inertia
3	Rectangular	R	R	1.75	1.81	1.80
3	Irregular	R	R	A	2.60	2.51
4	Rectangular	29.66*	29.44*	0.73	0.71	0.68
4	Irregular	R	R	1.16	1.06	1.14

Table 6: Fixed-reference parameter-to-geometry comparison. Only the CoM changes by +10 mm in x; force and endpoint differences are relative to full identification. Paired entries give original/shifted values. Geometry uses aircraft endpoints and a 0.27 m aircraft envelope.

Case	Force difference RMS (N)	Endpoint difference RMS (cm)	Min. endpoint distance (m)	Min. obstacle clearance (cm)
N3B	0.230	13.29	1.899/1.757	9.29/9.36
N3I	0.157	8.63	1.913/1.927	7.72/7.32
N4B	0.161	4.50	1.524/1.591	4.21/4.16
N4I	0.128	3.86	0.623/0.543	6.31/6.26

changes. This amplification includes active force and aperture constraints, rather than only an unconstrained wrench pseudoinverse. At the same reference, replacing central inertia by its mass-scaled uniform value while retaining identified mass and CoM changes endpoint RMS by less than 0.16 mm. These results independently locate the principal base-task benefit in CoM correction; they do not establish an additional tracking advantage from central-inertia identification.

A standard stagewise wrench allocator provides a separate force-planning comparison. At each of the same 81 time nodes it projects the outward preload and preceding force onto wrench equality, tension, vertical-span and nominal thrust constraints, without whole-interval derivative penalties or rate bounds. Both methods receive the full identified inertial model. After the same spline/stretch conversion, stagewise allocation has peak endpoint accelerations of 6.92–11.21 m/s^2, versus 0.80–1.99 m/s^2 for whole-interval allocation, and fails the common 2.25 m/s^2 candidate gate in all four frozen-reference cases. This comparison evaluates a specified standard allocation policy, rather than reproducing a complete published navigation system or asserting global infeasibility after retiming.

Multiplying all three derivative-penalty weights in Eq. (17) by 0.5 or 2 retains the sampled force, body-geometry and dynamic gates in all eight fixed-reference checks. Tension peaks are 7.31–8.17 N and endpoint acceleration peaks are 0.67–1.99 m/s^2. The coefficients therefore affect the distribution of feasible support positions; the complete candidate-level records retain these alternatives and their geometry differences. The 0.25–2 m residual-length scaling test in Section 3 yields 0.27–3.68 mm CoM errors and 5.06–8.80% central-inertia errors across the six nominal records.

Two additional identification baselines are executed through the same image-based controller and candidate search on the four-aircraft box and large-angle forked tasks, with sensor seed 28000. Raw unconstrained least squares produces a nonphysical central inertia in the box task and is rejected before transport; its forked execution completes with 3.68 cm peak position error and 2.17° peak attitude error. A support-free physically consistent baseline constrains the 4×4 pseudo-inertia matrix to be positive semidefinite [3], with the same 0.7–1.5 kg mass interval but no scanned spatial support. It completes both tasks: box/forked position peaks are 2.76/3.38 cm, attitude peaks are 0.67/2.16°, and tension ranges are 1.64–8.66/2.11–4.75 N. These results show that a physically consistent estimator without spatial support is also viable in the evaluated tasks. The proposed scan-supported model supplies a geometry-linked moment representation; the comparison does not establish a tracking advantage over the support-free PSD estimator. The baseline uses the same integral observation formulation and a 10^{-8} quadratic regularizer, and its full solver settings and original rejection record are archived.

5.6. Observation-record and support-model sensitivity

Two archived preparation records, the four-aircraft rectangular case and the large-angle forked case, are each refitted under four declared measurement perturbations and twenty seeds per perturbation. The 160 fits isolate added pose noise (2 mm translation and 0.15° rotation standard deviations), tension noise and bias (0.02 N each), fixed pose-frame bias (3 mm and 0.1°), and random observation removal (20%). These synthetic ranges define a record-space sensitivity analysis; they do not represent hardware calibration or newly executed occluded-image flights. For the forked case, CoM-error p95 is 4.76 mm with added pose noise, 5.50 mm with tension perturbation, 2.25 mm with frame bias and 0.89 mm with observation removal. The corresponding central-inertia error p95 values are 19.52%, 14.56%, 11.05% and 11.60%. The full distributions for both records are available with the analysis protocol.

Changing λ from 10^{-10} through 10^{-8} and 10^{-6} to 10^{-4} across all six nominal records changes CoM-error norms by at most 0.038 mm and central-inertia relative errors by at most 0.092 percentage points. A separate post-execution diagnostic restricts support locations to the evaluator's occupied body

material for the two forked records. Central-inertia errors become 5.85% and 6.72%, compared with 5.86% and 6.77% for the acquired-surface hull. This comparison tests the moment envelope under the same observed motion; evaluator material geometry is used only in this diagnostic.

5.7. Representative closed-loop measurement and actuator mismatch

The four-aircraft box and large-angle forked tasks are each executed under three perturbations, with sensor seed 28001 and fixed ballast: alternating tension-measurement biases of ±0.04 N, removal of every tenth delivered image observation, and an actual motor time constant of 37.5 ms instead of the nominal 25 ms. Planning and control retain the nominal motor model. These six new initial attempts use the process-isolated image pipeline and the unchanged execution gates. Table 7 separates initial planning, task completion and full acceptance. Five attempts obtain a plan and complete transport; three satisfy every acceptance criterion. The two observation-loss tasks pass, with box/forked attitude peaks of 0.73/4.70° and minimum aircraft-envelope clearances of 0.0409/0.0811 m.

The tension-biased box execution passes, whereas the forked execution reaches 8.63° attitude error with 5.02 mm CoM estimation error. With the slower motors, the box completes but reaches 7.92° payload attitude error, 45.87° aircraft tilt and 6.90 m/s^2 aircraft acceleration. The forked attempt develops a preparation transient, including cable unloading, and obtains no accepted course candidate. Thus the tests distinguish tolerance to the specified observation removal from sensitivity to unmodeled actuator lag and tension bias during large rotations. The sample size describes these particular mismatch conditions; it is not a population robustness estimate. Per-attempt position errors, planning diagnostics, contacts, tensions and clearances are public with the original records.

5.8. Sequential apertures and large-angle reorientation

The course comprises a 28 m centerline and twelve oriented solid obstacles: a laterally offset narrow frame, a low ceiling, a tilted frame, and three staggered blocks along the approach and exit regions. Figure 7 shows the actual load and aircraft trajectories relative to these obstacles. Two course executions use the same forked object and corner mounts. The moderate-rotation course has a 2.70 m first opening and an 85 s transport. The large-rotation course narrows this opening to 2.10 m and uses a 125 s transport. The ceiling underside is 4.45 m above the ground, and the tilted opening has 2.85 m clear width.

In the large-rotation execution, the load is nearly horizontal at the start, rotates side-on for the first frame, returns to horizontal while changing heading under the ceiling, and rolls toward the opposite side for the last opening. Figure 1(a)–(c) displays the three crossing configurations. The actual roll ranges from −86.64° to 90.34°, and yaw ranges from −25.25° to 90.85°. At the three crossing instants the roll angles are 90.03°, 0.03° and −84.24°. These rotations are achieved through cable loading while the team progresses along the course.

Figure 8 relates these attitudes to the supporting formation. The maximum vertical aircraft span is 1.37 m; it falls to approximately 0.19 m at the low-ceiling crossing. The minimum inter-aircraft separation over the complete execution is 0.640 m. The formation is consequently layered during side-on support and flattened under the ceiling. Payload attitude, relative aircraft position and available space change together.

Table 8 reports the course results. The large-rotation execution has a 3.34 cm peak position error and 2.19° peak attitude error. Tensions remain between 2.11 and 4.75 N, maximum aircraft tilt is 16.13°, and maximum aircraft speed is 2.38 m/s. The actual payload peak speed is 1.05 m/s. There are no rigid-body contacts, sampled cable-obstacle or cable-body intersections, slack cables or motor saturation intervals. All 1682 processed visual observations are valid under the registration tests.

The large-rotation geometric audit finds 0.081 m minimum aircraft clearance after a 0.28 m envelope inflation and 0.268 m minimum scanned-surface clearance after 0.045 m inflation. Complete cable segments are checked with a 0.0468 m radius including termination width and observed half-sample motion. These are sampled geometric margins with the stated motion inflation; they are not a general continuous-time safety certificate under arbitrary execution disturbances. A finer geometric audit replays four two-second command intervals covering the closest aircraft approach and all three aperture crossings, using 0.5 ms integration and 2 ms geometry samples without interior resets. The minimum aircraft clearance is 0.081088 m, versus 0.081089 m at 20 ms in the same replay, and no inflated cable segment intersects an obstacle. Interval speed maxima and half-sample translation scales accompany the records. These checks refine the reported execution margins without asserting disturbance-invariant safety. The corner attachment geometry remains outside the material body during rotation. The full physical task takes 169.58 s, including 16 s preparation, identification and planning waits, transport, and terminal holding.

5.9. Computation and numerical cross-checks

For the four base identified-model configurations, initial whole-segment planning takes 0.74–1.56 s. Image processing p95 is 39.7–41.3 ms, control computation p95 is 1.55–1.67 ms, and sensor age p95 is 120–140 ms. The larger course requires additional conic separation updates and full-surface checks. Its initial plan takes 24.85 s, image processing p95 is 91.03 ms and control computation p95 is 4.57 ms. The observations remain asynchronous, with sensor age p95 of 160 ms. These timings separate pretransport trajectory construction from visual processing and feedback during motion. Force planning uses Clarabel with feasibility tolerance 2×10^{-8}, absolute and relative gap tolerances 10^{-7}, and at most 100 iterations. Moment fitting uses feasibility tolerance 10^{-9} and absolute gap tolerance 10^{-10}. The archived solver diagnostics retain the iteration counts for individual candidates.

Numerical consistency is examined using a two-second interval from the three-aircraft irregular-payload base execution, starting at physical time 29.30 s. A separately coded NumPy rigid-body model integrates the recorded motor inputs

Table 7: Representative perturbed closed-loop attempts, one execution per condition. P/C/A denotes plan obtained / transport completed / all gates accepted (Y or N). Attitude peaks concern transport; tension extrema include preparation. Clearance is the minimum sampled aircraft gap after 0.28 m inflation. Dashes denote no transport.

Task	Perturbation	P/C/A	Peak attitude (°)	Tension (N)	Min. clearance (cm)
Box	Tension bias	Y/Y/Y	0.90	1.64–8.62	4.10
Box	Motor lag	Y/Y/N	7.92	1.06–8.74	7.05
Box	Observation loss	Y/Y/Y	0.73	1.64–8.66	4.09
Fork	Tension bias	Y/Y/N	8.63	2.11–4.80	8.02
Fork	Motor lag	N/N/N	–	0.00–5.61	–
Fork	Observation loss	Y/Y/Y	4.70	2.11–4.76	8.11

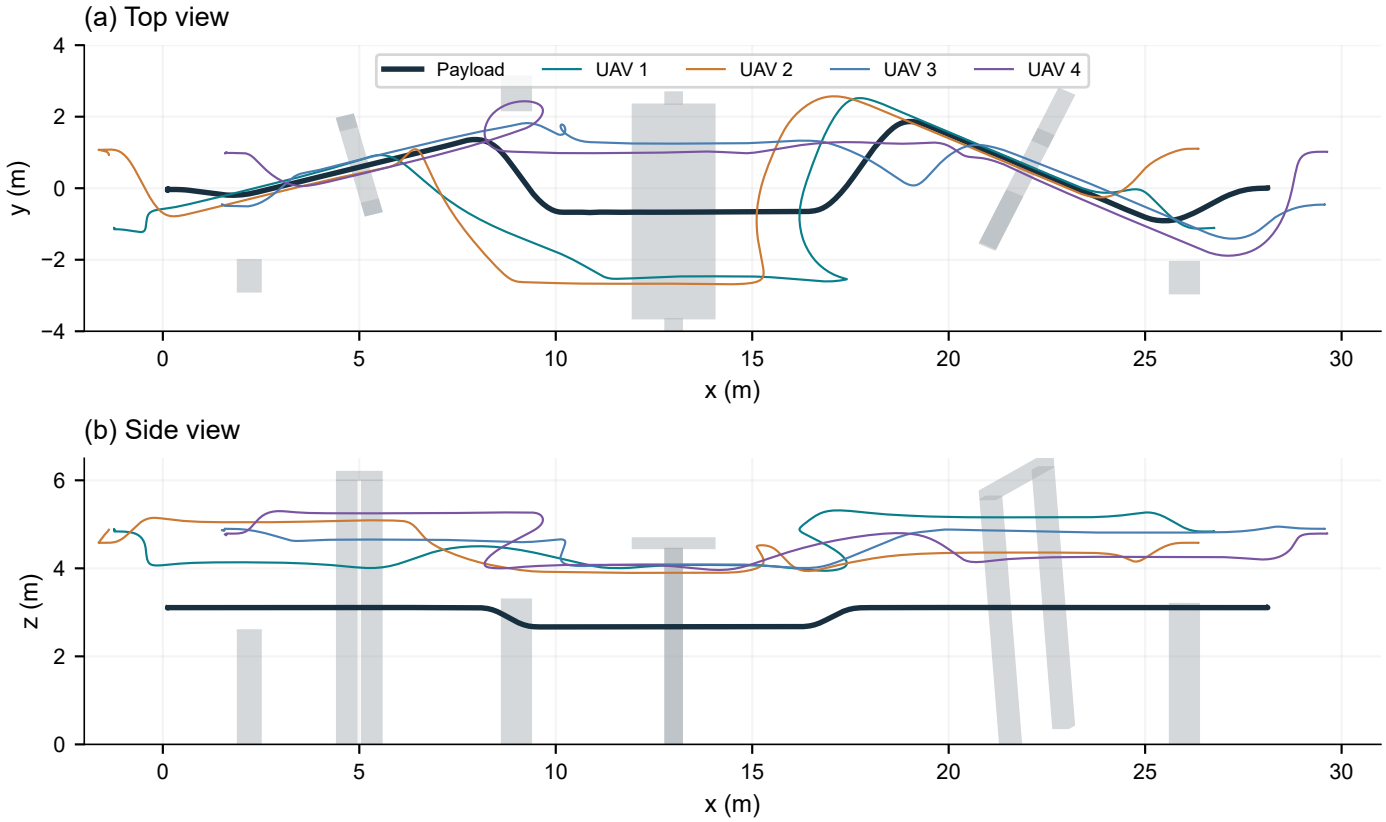


Figure 7: Actual large-rotation course trajectories in (a) top and (b) side views. Gray polygons are projections of the twelve oriented obstacle boxes. The payload centerline and all four aircraft centers are plotted from the same dynamic execution.

Table 8: Forked-payload course executions. The two tasks have different first-aperture widths and prescribed rotation amplitudes; each is evaluated against the common execution criteria.

Course	Transport (s)	Peak position (cm)	Peak attitude (°)	Tension (N)	Min. separation (m)	Planning (s)
Moderate rotation	85	4.06	1.73	2.11–4.89	0.946	14.69
Large rotation	125	3.34	2.19	2.11–4.75	0.640	24.85

with fourth-order Runge–Kutta at 0.25 ms, without resetting the state inside the interval. As the MuJoCo step decreases from 1 to 0.5 and 0.25 ms, the maximum payload CoM difference decreases from 0.973 to 0.484 and 0.242 mm. The corresponding rotation differences are 0.1291°, 0.0642° and 0.0320°; aircraft CoM differences decrease from 3.468 to 1.727 and 0.862 mm. Figure 9(a),(b) includes this interval together with three additional two-second intervals from the large-angle course: maximum absolute roll, maximum sampled tension variation, and minimum aircraft-envelope clearance. Each interval is replayed continuously without interior state resets. At a 0.5 ms MuJoCo step, the added intervals differ from the independent RK4 replay by at most 0.365 mm in payload CoM and 0.040° in attitude; halving the step approximately halves these differences.

A complete closed-loop rerun of that base configuration at

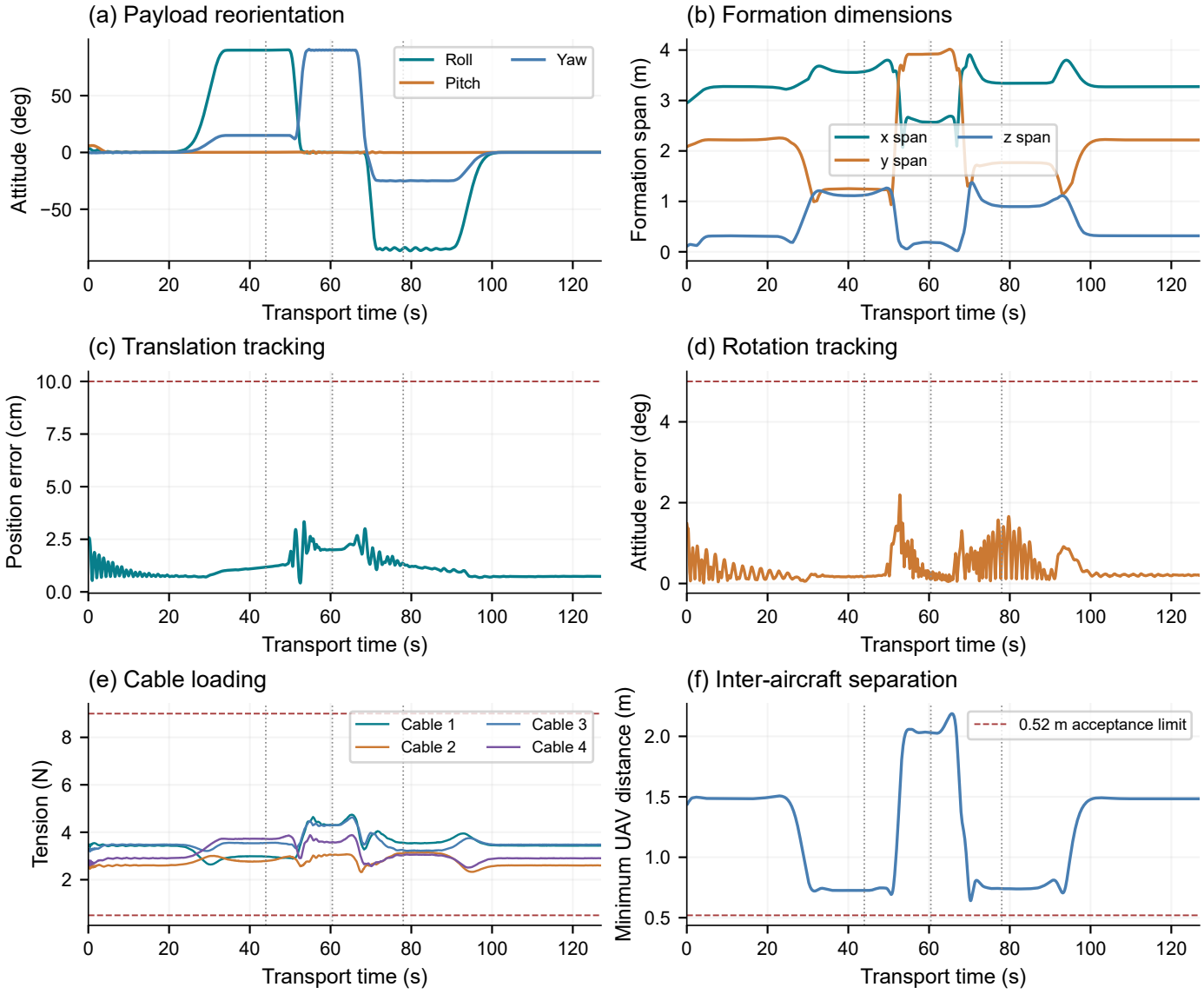


Figure 8: Large-rotation course dynamics. (a) Payload attitude. (b) Aircraft center span along each axis. (c),(d) Position and rotation errors. (e) Cable tensions. (f) Minimum pairwise aircraft center distance. Dotted vertical lines indicate the three payload crossing instants. Dashed horizontal lines indicate the corresponding execution or separation thresholds.

a 0.25 ms step also satisfies the execution criteria, with 3.53 cm peak position error, 2.59° peak attitude error and 1.20–7.53 N cable tensions. Together, the interval comparison and refined closed-loop execution examine both the numerical force implementation and the resulting feedback trajectory.

5.10. Implications for cooperative transport

The experiments connect three different kinds of information. The scan supplies the object's occupied geometry and a consistent frame for calibrated attachments. Tension and motion reveal the mass moments needed to balance and rotate that object. The force-to-endpoint map then converts the identified mechanics into aircraft geometry. The paired base experiment quantifies the effect of supplying those moments, while the sequential-aperture experiment shows how compliant support geometry changes during substantial load rotation.

The evaluated configuration uses calibrated external virtual RGB-D views, static payload mass distributions and known obstacle geometry. These conditions define the sensing and task model of the reported experiments. An onboard implementation introduces camera occlusion, calibration drift and different image coverage; changing payload contents introduces time-varying inertial parameters. These extensions call for corresponding sensing and replanning mechanisms. Within the evaluated setting, the current results establish continuous large-angle closed-loop transport across the specified apertures, with explicit load, cable and aircraft measurements.

6. Conclusion

This paper develops a vision- and tension-informed connection between inertial identification and supporting formation ge-

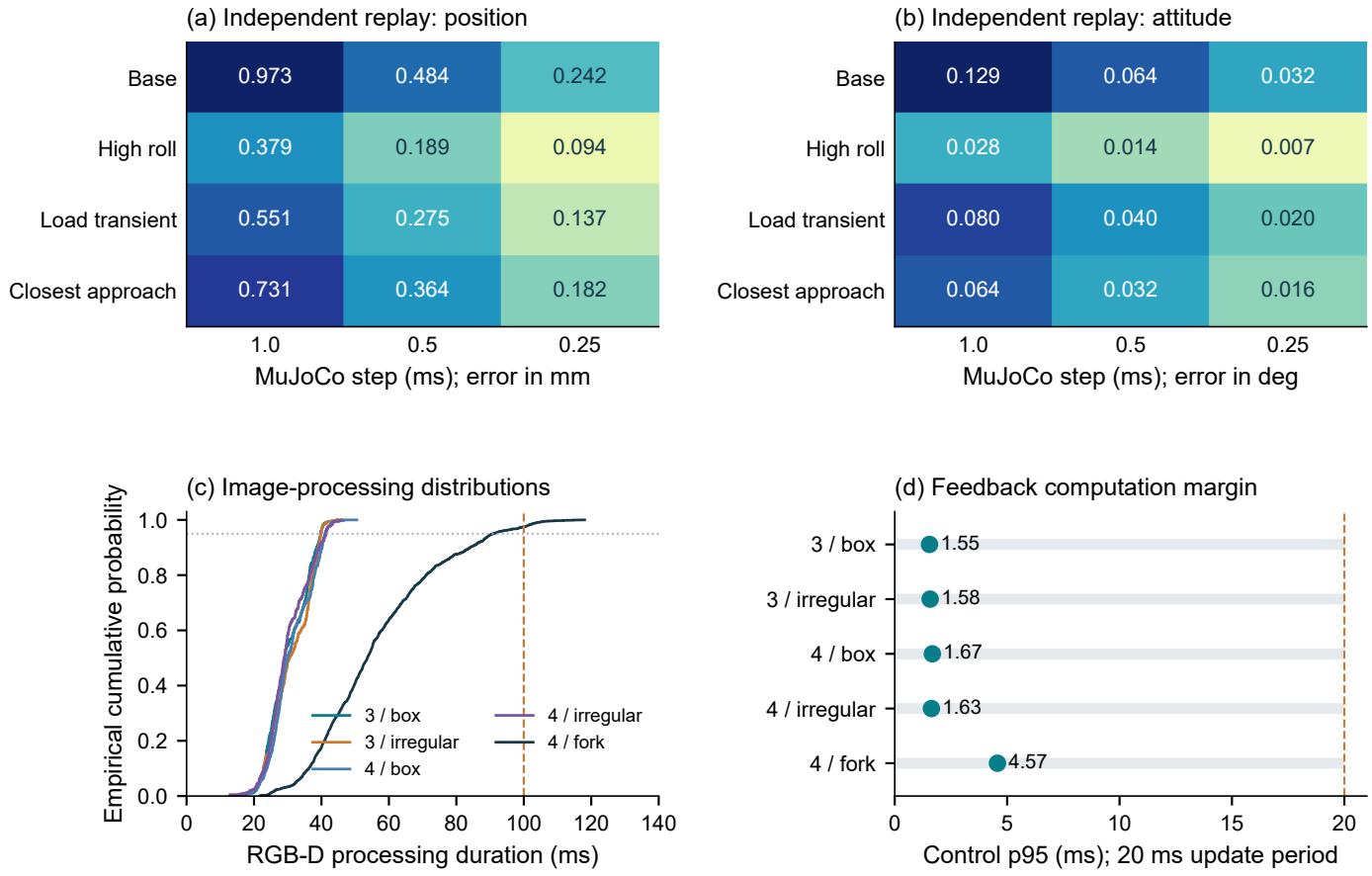


Figure 9: Numerical verification and computation distributions. (a),(b) Maximum payload CoM and attitude differences from independent two-second RK4 motor-command replay at three MuJoCo steps, including the base interval and three mechanically demanding course intervals. (c) Empirical cumulative distributions from all archived image-processing durations; the dashed vertical line marks the 100 ms RGB-D request interval. (d) Control-computation p95 values relative to the 20 ms control period. The separate computation paths do not imply 50 Hz image registration or whole-segment replanning.

ometry for cooperative aerial transport through known obstacles. The payload pose reference family is prescribed; conic optimization selects the cable forces, compliant support endpoints and a feasible timing candidate. A reconstructed surface anchors markerless pose measurements and a nonnegative representation of inertial moments. Integral momentum observations identify the CoM and inertia used by a conic cable-force planner. Compliant endpoint construction and sequential separation then connect these forces to the aircraft formation throughout rotation and aperture passage.

Three-aircraft and four-aircraft rectangular/irregular configurations complete the base task with 2.62–4.10 cm position errors and 0.66–2.64° attitude errors. The mass-matched five-arm comparison attributes the main base-task tracking gain to CoM correction, while complete-inertia accuracy and transport-only prediction quantify the additional identified moments. A forked payload subsequently completes a 28 m course with successive side-on, horizontal and opposite-side orientations, reaching approximately ±90° roll while maintaining positive cable tensions and clearance. A fixed-reference sensitivity test directly traces CoM changes into cable forces and supporting endpoint positions. Process isolation also increases the observed completion rate from 16/20 to 20/20 on the same twenty sensor seeds, with all original attempts retained. These findings support the use of identified internal mass moments when coordinating payload attitude and supporting formation through spatial restrictions.

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Data availability

The simulation and identification source code, task configurations, true and fitted inertial parameters, archived trajectories, ablation and repeated-seed experiments and figure-generation scripts are available at `https://github.com/shepherd-p/ForestSling`, with the manuscript snapshot fixed at tag `v0.2.0`. The repository documents the analysis and rerun commands and identifies each experiment by its configuration and random seed. Recorded trajectory visualizations are provided with the reproducibility materials. Proprietary Unreal

environment assets are excluded; numerical analysis and MuJoCo simulation do not require those assets.

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